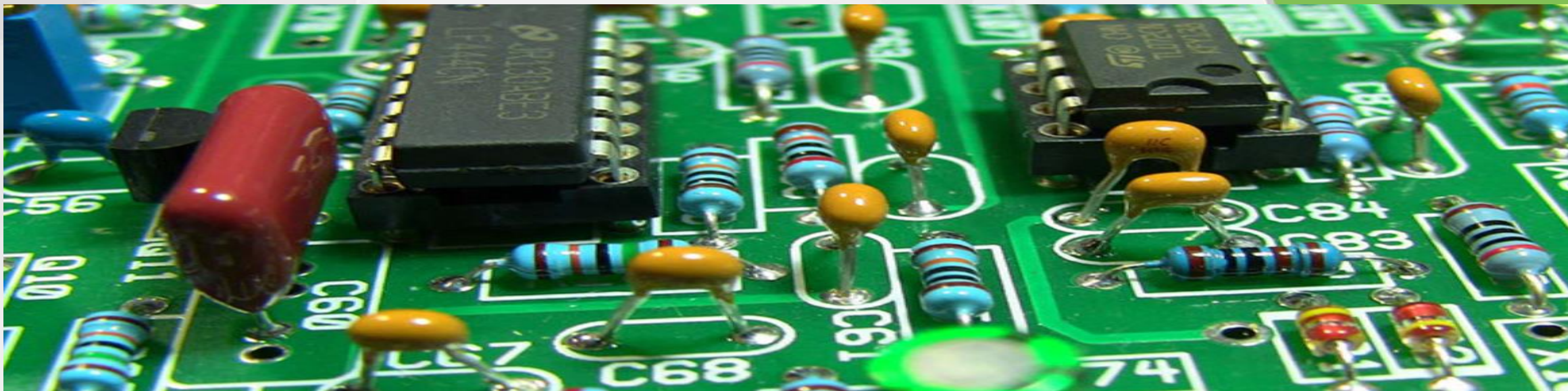


34621
Electromagnetic sensors and digital signal processing

Technical University of Denmark

Course content / learning objectives

Kursusinformation

Engelsk titel	Electromagnetic sensors and digital signal processing
Undervisningssprog	Dansk
Point(ECTS)	5
Kursustype	Diplomingeniør Obligatorisk kursus (B Eng), Elektroteknologi
Skemaplacering	Januar
Undervisningens placering	Campus Lyngby Campus Lyngby og Campus Ballerup
Undervisningsform	Praktisk design med høj grad af selv-motivering. Der arbejdes i grupper af 2-3 studerende.
Kursets varighed	3-uger
Evalueringsform	Bedømmelse af opgave(r)/rapport(er) Nogle dage er planlagt med obligatorisk aktivitet, f.eks. forelæsning, præsentation og demonstration for andre grupper mv. Til slut af afleveres rapport. Karakteren baseres på et helhedsindtryk.
Bedømmelsesform	bestået/ikke bestået , ekstern censur
Tidligere kursus	62739
Pointspærring	62739
Faglige forudsætninger	346013003562743 eller tilsvarende , Viden om elektriske og magnetiske felter (svarende til del-elemeter i kursus 30035 eller 62774) samt grundlæggende elektronik inkl. forstærker design. Viden om lineær og digital signalbehandling incl filterdesign, spectrum analyser etc. Den studerende skal være i besiddelse af og fortrolig med anvendelse af eget processor-kit fra kursets start - f.eks. fra tidligere kursus i digital teknik og programmering.
Kursusansvarlig	Per Lynggaard , Lyngby Campus, Bygning 325, Tlf. (+45) 4525 3486 , plyn@dtu.dk
Institut	34 Institut for Elektroteknologi og Fotonik
Tilmelding	I studieplanlæggeren

Overordnede kursusmål

At sætte den studerende i stand til at
- specificere og designe et elektronisk sensor system indeholdende en eller flere sensorer
- implementere og anvende et elektronisk sensor system, hvor signaler fra sensorer efterbehandles i et digitalt processor system

Læringsmål

En studerende, der fuldt ud har opfyldt kursets mål, vil kunne:

- Anvende begreber, teori og metoder indenfor det elektromagnetiske emneområde såsom: feltstørrelserne i magnetostatik, Biot-Savarts lov, Gauss' lov, magnetiske felter omkring ledere og de tilsvarende grænsebetingelser (svarende til del-elemeter i kursus 30035)
- Anvende begreber, teori og metoder indenfor digital signal behandlings emneområdet, såsom: nødvendig samplingsfrekvens,anti-aliasing filter, diskret Fourier transformation og design af digitale filtre (svarende til del-elementer i kursus 62743)
- Opstille, beskrive, udvikle og anvende modeller for de metoder og komponenter, som anvendes i kurset.
- Specificere, designe, implementere, teste og dokumentere digital signal processing applikationer og programmer
- Specificere, designe, beregne, analysere, modellere, simulere, konstruere, teste og dokumentere hardware
- Samarbejde i grupper om løsning af fælles opgaver
- Planlægge, specificere, samarbejde, dokumentere and anvende relevant ingeniørmæssige terminologi og sprogbrug.
- Relatere mellem specifikationer, beregninger, simuleringer og målinger samt drage klare og revelante konklusioner
- Begrunde årsager og forskelle mellem beregnede, simulerede og målte resultater.
- Diskutere forskelle i forhold til relevant teori, simulationer og tests.
- Formidle gruppens planlægningsproces og opnåede mål

Kursusindhold

Der arbejdes ud fra en bunden opgave (udleveres ved kursets start).
Der opstilles en endelig specifikation og designet implementeres vha et digitalt signalbehandlingssystem indeholdende en eller flere sensorer f.eks. i form af et mikroprocessor system (gerne en dedikeret dsp) med tilhørende elektromagnetiske/kapacitive/trådløse sensorer.
Sensorernes fysiske opbygning og virkemåde beskrives, analyseres og dokumenteres.
Signalbehandlingssystemets virkemåde beskrives, analyseres og dokumenteres.
Den samlede løsning dokumenteres.

Elektronikbaserede systemer er oftest en primær teknologi-driver i bæredygtighedsudviklingen, hvor disse bidrager til logistikhåndtering af bæredygtige ressourcer i en social, en økonomisk og en miljømæssig dimension. Denne bæredygtighedsvinkel medtages implicit i kursets læringsmål.

Bemærkninger

Elektroteknologi: 3. semester

Der arbejdes i mindre grupper

Practical information's

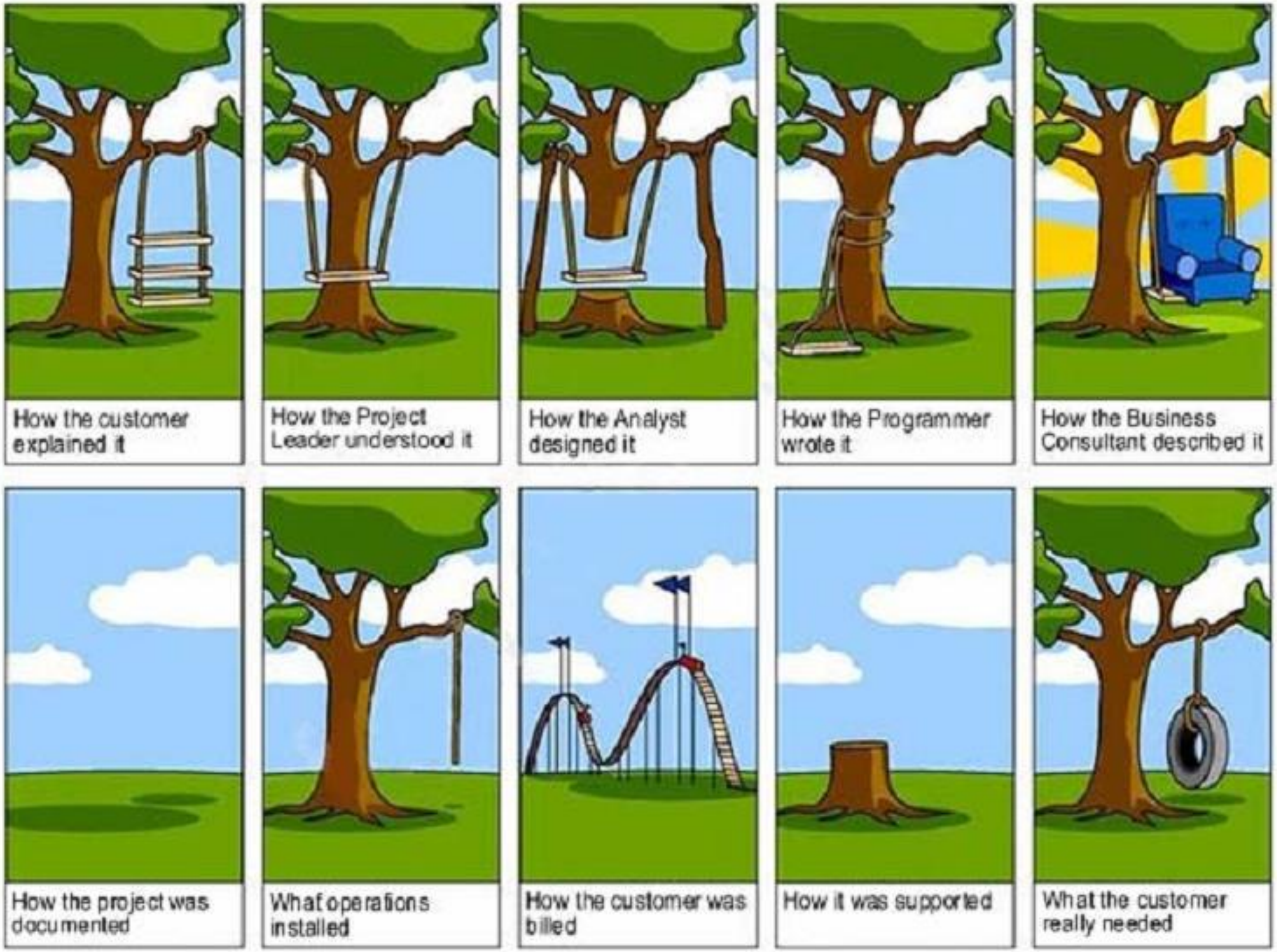
- The course is running from 5/1-2026 to 23/1-2026.
- Plan (next slide)
- The first day (today) we will start with a technical introduction to the problem and look at some ideas to solve the most complex parts of the project.
- We are two supervisors Tjalfe and me.
- Room X1.28 has been reserved. So, X1.28 can be use as a meeting room for group meetings, etc.
- Furthermore, we have reserved
  - Ballerup Campus V1.01 and V1.04 from 8.00 to 18.00;
- Necessary wire/probe-kits will be available from F1.01.
- Components etc. can be found in the component-shop / mechanical lab.
- Groups formation – each group must hand-in 1 page with names, study number, and pictures - ***Must be uploaded today (Learn- mycourse/groups)***

Course plan

	Kursusplan 34621, Jan. 2026														
	05-jan	06-jan	07-jan	08-jan	09-jan	12-jan	13-jan	14-jan	15-jan	16-jan	19-jan	20-jan	21-jan	22-jan	23-jan
	M	T	O	T	F	M	T	O	T	F	M	T	O	T	F
Kursus introduktion	10-12														
gruppedannelse	13-14														
Projekt planlægning / krav orientering	14-16														
biplor transistor kursus		13-14													
Kursusafrunding															14-15
Lokation (Lab=V1.01+V1.04)	X1.28	X1.28/Lab	Lab	Lab	Lab	Lab	Lab	Lab	Lab	Lab	Lab	Lab	Lab	Lab	X1.28
Hjælpelære (Tjalfe)	9-15	9-15	9-15	9-15	9-15	9-15	9-15	9-15		9-15	9-15	9-15	9-15	9-15	9-15
Per Ballerup	10-16	10-15	10-15	10-15			10-15			10-15	10-15			10-15	10-16

Align expectations

What's the problem in most projects ?



Why using structured methods anyway?

Non Structured | Structured

-
- **Start:**

- **30% of time used:**

- **70% of time used:**

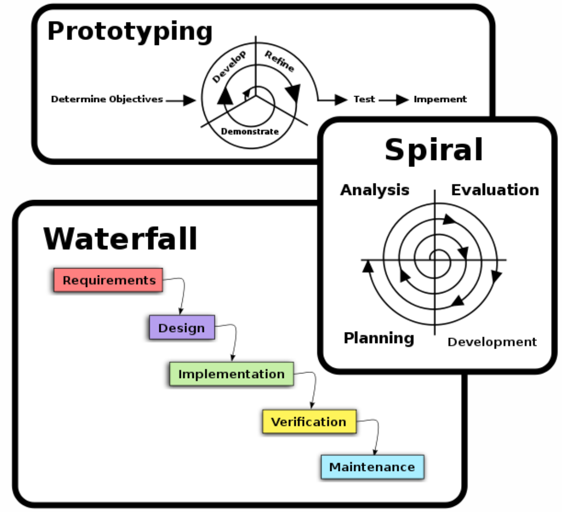
- **100% of time used:**

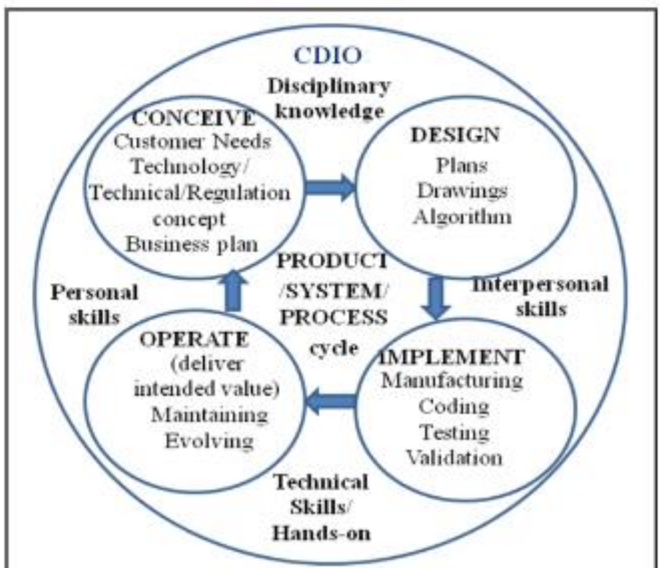
- **200% of time used:**

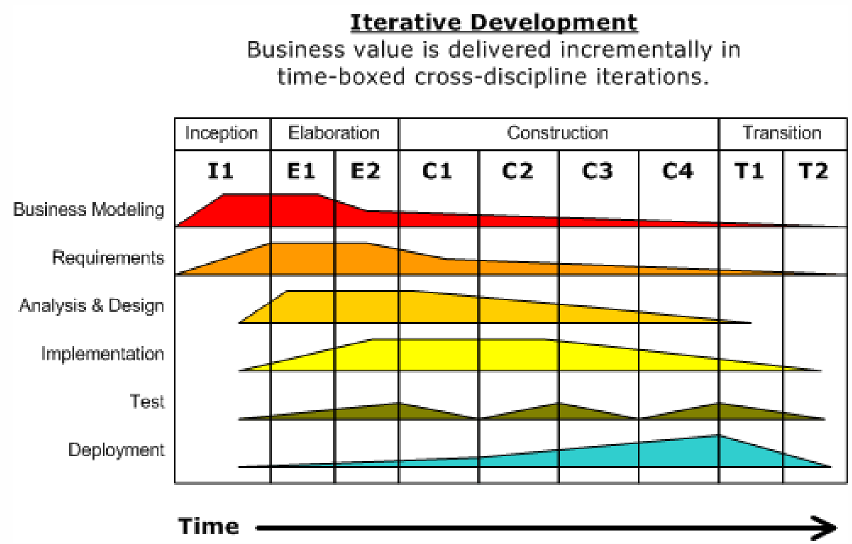
Keywords:
Maintenance
Document Presentation
time Management
Risk Minimization
Quality Optimization
Reuse
etc.

Development methods

- **Many agile methods exist, but common for them is they make use of common elements such as:**
  - requirement specifications
  - design specifications
  - verification strategies.
- **Example: Rational Unified Process (RUP)**
- **CDIO**

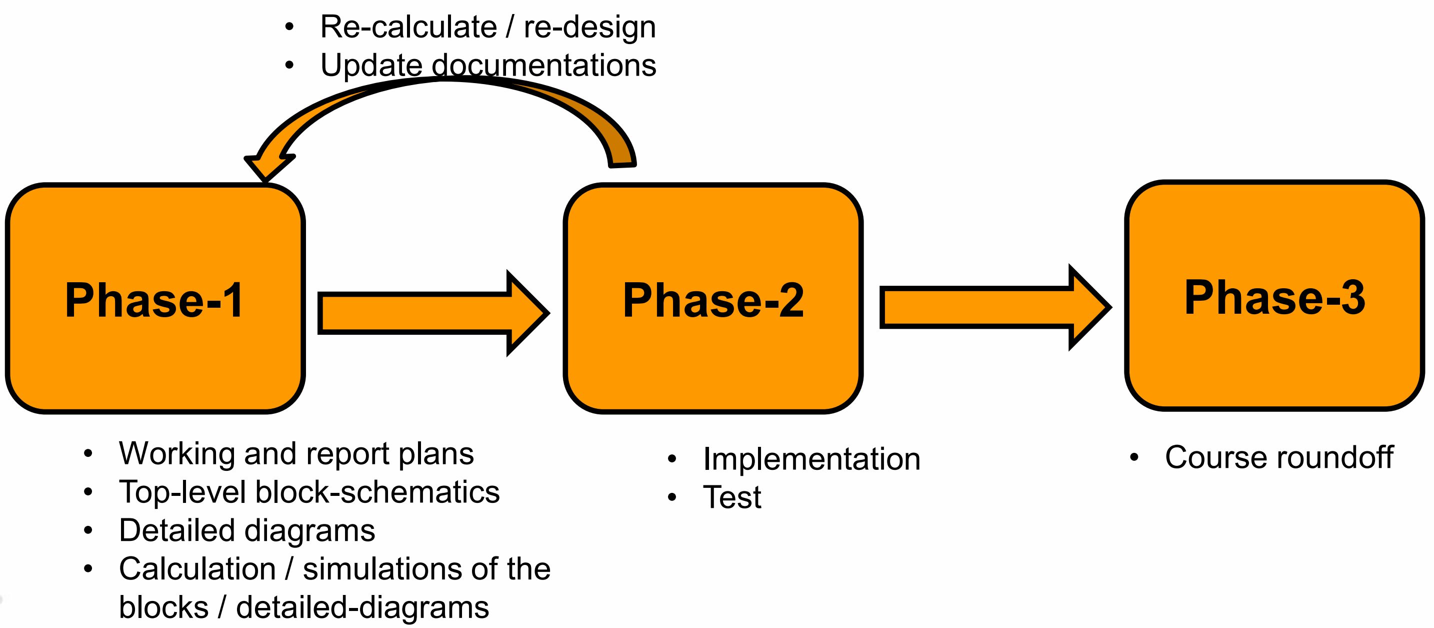






Project plan

- **The project is divided into 3-phases:**
  - Phase-1: see plan for this phase (Arbejdsplan) – SOTA, analysis, design, implementation.
  - Phase-2: implement the metal-detector in lab.
  - Phase-3: course roundoff.



Method

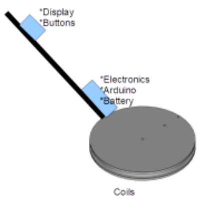
- **Læs kravspecifikationen og diskuter den i gruppen – det er vigtigt at forstå kravene**
- **Fremstil en plan for 3-ugers forløbet – hvem laver hvad, hvornår (f.eks. et Gantt-kort), planen skal som minimum indeholde faserne: SOTA, Analyse, Design og Implementering (se nedenfor) – *planen skal inkluderes i rapporten som bilag*.**
- **State-Of-The-Art / SOTA (dvs. et forstudie) hvor tilgængelig litteratur læses, forstås og diskuteres. Inkluder andre kilder (vær kritisk) såsom hjemmesider, bøger og specielt viden fra tidligere kurser, m.m.**
- **Analyse: (generelt et af de vigtigste afsnit i en rapport)**
  - Baseret på SOTA skal mulige og alternativer løsninger til de stillede krav diskuteres og den mest optimale løsning vælges – HUSK der skal argumenteres for disse valg med gode argumenter - forklar hvorledes du kommer frem til denne løsning og hvorfor ikke en af de andre løsninger.
  - Check løsningen opfylder alle kravene.
  - Fremstil et overordnet blokdiagram / principdiagram over metaldetektorens elementer – brug blokdiagrammet fra opgaveteksten som udgangspunkt – *blokdiagrammet skal inkluderes i rapporten.*
- **Design: (også et vigtigt afsnit i en rapport)**
  - Fremstil et (elektrisk) design af de enkelte blokke i blokdiagrammet fra analysefasen.
  - Design valg, hvorfor dette design – alternativer? **(husk at bruge resultaterne fra analyseafsnittet og kravene fra kravspecifikationen**)
  - Beregn og simuler de elektriske kredsløb i f.eks. LT-Spice.
  - Fremstil diagrammer – husk at forklare hvorfor det ser ud som det gør - inkluder beregnede værdier.
  - Husk at fremhæve de ingeniørmæssige bidrag til løsningen.
- **Implementering og test:**
  - Fremstil de designede kredsløb i laboratoriet på Campus Ballerup og få dem til at virke. Dvs. spoler, forstærkere og software, m.m.
  - Planlæg hvorledes apparatet testes – ***brug kravene fra kravspecifikationen som input.***
  - Fortag testene, beskriv testresultaterne og sammenlign med forventede værdier

Overview – requirement specification

- **Requirements specification – see document (kravspecifikation.pdf)**

KRAVSPECIFIKATION FOR UDVIKLING AF METALDETEKTOR I 3-UGERS KURSET F2021

Der skal udvikles en metaldetektor som opfylder nedenstående krav. Et eksempel på implantation er vist nedenfor (kun et eksempel – IKKE et krav at detektoren skal udformes som vist).

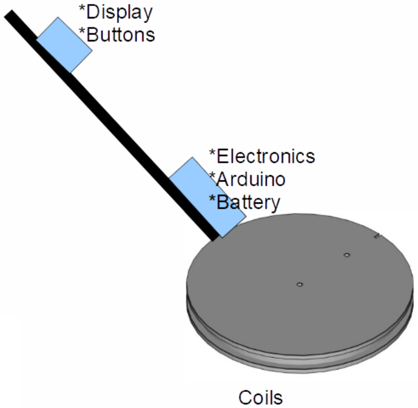


Nummer:	Navn:	Beskrivelse:	Prioritet:
1	Amplitude / fase detektion	Det skal være muligt at kunne detektere amplitude og fase i det reflekterede signal	1
2	Metal type	Skal kunne skelne jern fra kobber, messing og aluminium	1
3	Distance krav	Skal kunne detektere en jerngenstand (radius på ca. 15 mm. og en længde på 50 mm) i en dybde på 50 mm. i fri luft.	1
4	Strømforsyning	Hele metaldetektoren skal kunne køre på et 9 volts batteri (batteristørrelse E / 6LR61) – kun et batteri.	1
5	Funktionstid	Metaldetektoren skal kunne køre minimum 100 minutter på batteriet og den ubelastede rest-spænding skal være større end 6 volt efter de 100 minutter.	1
6a	Processor design	Kredsløbet skal være baseret på en Arduno	1
6b	Hardware design	Kredsløbene til at drive spolerne og forstærke det modtagne signal skal bygges med diskrete komponenter såsom: transistorer, operationsforstærkere, m.m.	1
7	Muligt software design	Brug Arduinoens ADC og kontroller den med timer interrupts	2
8a	Bruger interface display	Skal kunne udskrive amplitude og fase i et display	1
8b	Display udlæsningsstabilitet	Værdierne i displayet skal være læsbar for et flertal af brugerne (spørg om hjælp)	1
8c	Forbedret display udlæsningsstabilitet	Der anvendes et FIR elle IIR filter til at lavpas-filtrere (midle) udlæsningen.	2
9a	Bruger interface primære knapper	Der skal være en start/stop knap	1
9b	Bruger interface sekundære knapper	Der skal være en nulstillingsknap (fratrækker værdierne når detektoren ikke er i nærheden af metal)	2
10	Detektions princip	Very Low Frequency (VLF)	1
11	Samplingsfrekvens	8 kHz +/- 100 Hz	1
12	Detektionsfrekvens	2 kHz +/- 100 Hz	1
13	Metaldetektorudformning	3D- printet / trækonstruktion	2
14	Spolernes bæreenhed	3D- printet spoleform / læserskåret plastik	2
15	Spoleudformning	Spolerne anvender en centreret udforming med sende-spolen yderst og feedback / modtager-spolen inderst.	2

Prioritet:

1: Skal designes og implementeres

2: Frivilligt om disse krav medtages.



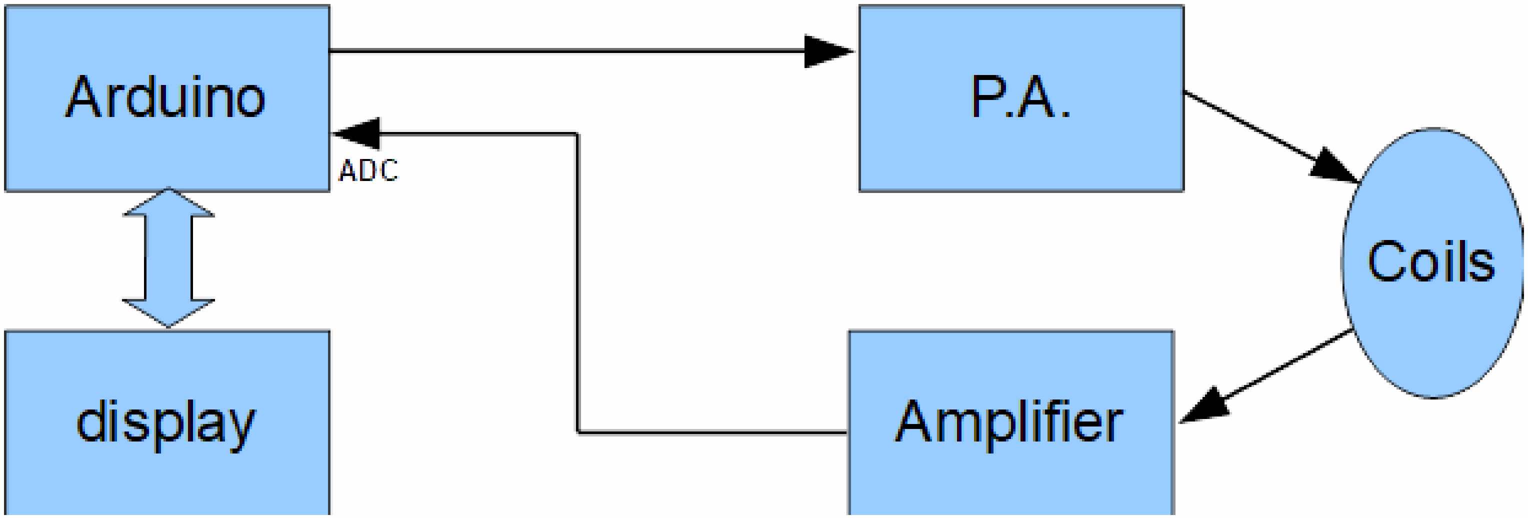
An example, i.e. - only an example, feel free to do your own design …

Report requirements

- **Must be hand-in before 23/1-2026 at midnight.**
- **The report must be structured as:**
  - Front page with group number, names and pictures of the members in the group.
  - Table of content
  - Short introduction (0.5-1 page)
  - Main sections (analyse, design, implementation and test).
  - Short conclusion including reflection over: to what extent are the requirements fulfilled, what could have been done differently, reflections over the group work (0.5-1 page).
  - ***A description / table of the contributions made by each group-member.***
  - Appendix – group plan(e.g. Gantt chart) + …
- **The report size is limited to max. 20 pages (+some extra pages for appendixes, if needed).**
- **The report must contain calculations, drawings, and elaborations as required in the followings:**
  - Commented DSP code, calculations, and implementation (Arduino code)
  - Calculations for all coils, including comparison with the realized ones – elaborate differences
  - LT-Spice models covering the P.A. and the Rx-coil amplifier, including short explanations and elaborations of design decisions.
  - Elaborated discussion of the measured results in relation to the relevant theories.
- **Conclusion: elaborate how, and to what degree, the requirements are fulfilled.**

Overview

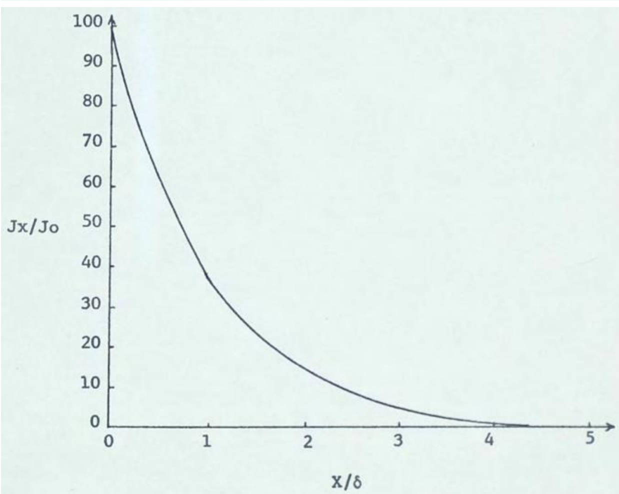
- **A schematics covering the data processing part of the exercise :**

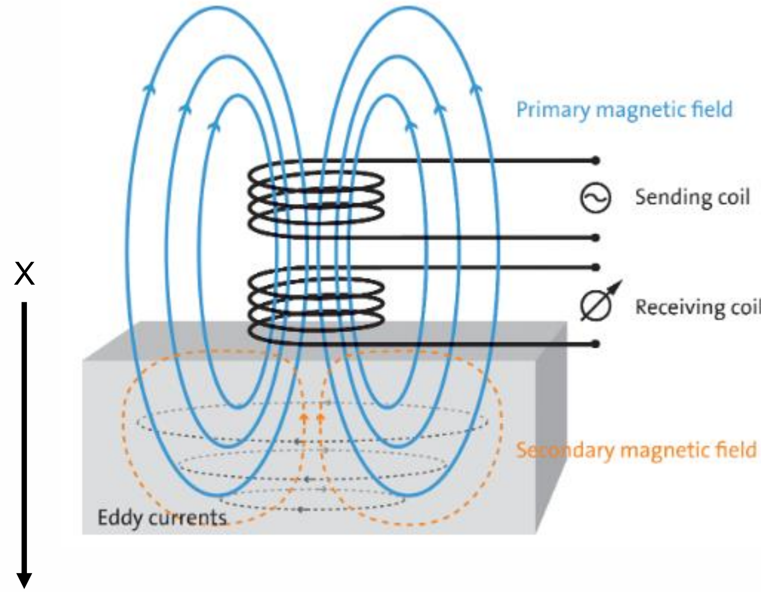


VLF M.D. working principle

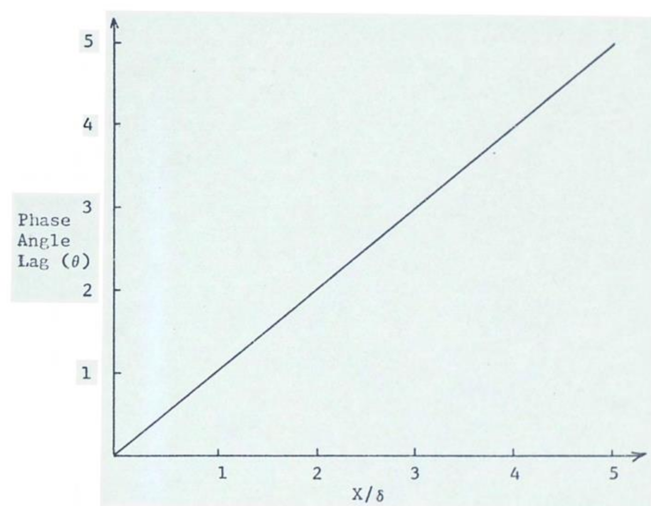
- **The depth of eddy-currents is given by σ, μ, and ω.**

$$\hat{J}y(x) = \hat{J}y(0)\exp\left[-(1 + j)\left(\frac{\omega\mu\sigma}{2}\right)^{\frac{1}{2}}x\right]$$





$$\delta = \left(\frac{2}{\omega\mu\sigma}\right)^{\frac{1}{2}}$$



δ depth where current is reduced to 37%
σ conductivity
ω angular frequency
μ permeability
X penetration depth

Theory of Eddy Currents for Non-destructive Testing, 1976, Craig Charles Biddle

VLF M.D. working principle

- **Examples of permeability and conductivity of common materials:**

Material	Conductivity σ (S/m) at 20 °C
Silver[d]	6.30×10^7
Copper[e]	5.96×10^7
Annealed copper[f]	5.80×10^7
Gold[g]	4.11×10^7
Aluminium[h]	3.77×10^7
Calcium	2.98×10^7
Tungsten	1.79×10^7
Zinc	1.69×10^7
Nickel	1.43×10^7
Lithium	1.08×10^7
Iron	10^7

Medium	Susceptibility, Permeability, μ (H/m)
Metglas 2714A (annealed)	1.26×10^0
Iron (99.95% pure Fe annealed in H)	2.5×10^{-1}
Iron (99.8% pure)	6.3×10^{-3}
Aluminum	$1.256\,665 \times 10^{-6}$
Wood	$1.256\,637\,60 \times 10^{-6}$
Air	$1.256\,637\,53 \times 10^{-6}$
Concrete (dry)	
Vacuum	$4\pi \times 10^{-7}$ (μ_0)
Hydrogen	$1.256\,6371 \times 10^{-6}$
Teflon	1.2567×10^{-6}[15]
Sapphire	$1.256\,6368 \times 10^{-6}$
Copper	$1.256\,629 \times 10^{-6}$

- **Example copper:**

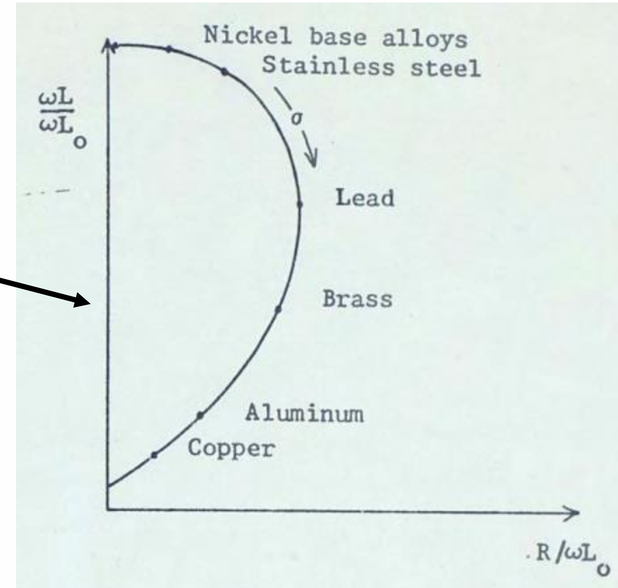
$\omega := 2 \cdot \pi \cdot 8e3 :$
$\mu := 1.256e-6 :$
$\sigma := 5.96e7 :$

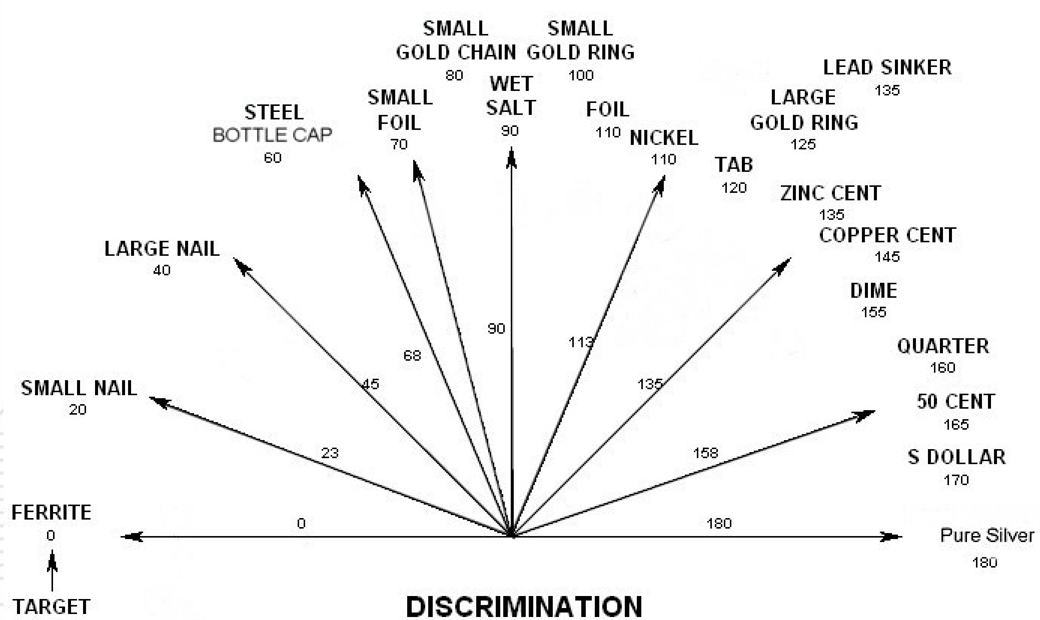
$$\delta := \sqrt{\frac{2}{\omega \cdot \mu \cdot \sigma}}$$

$\delta := 0.0007290580685$

VLF M.D. working principle

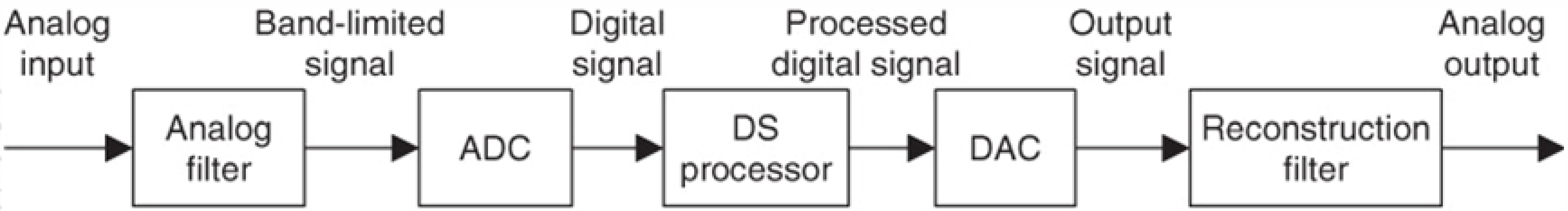
- **Receiver coil current at x-axis and loaded coil voltage at y-axis**





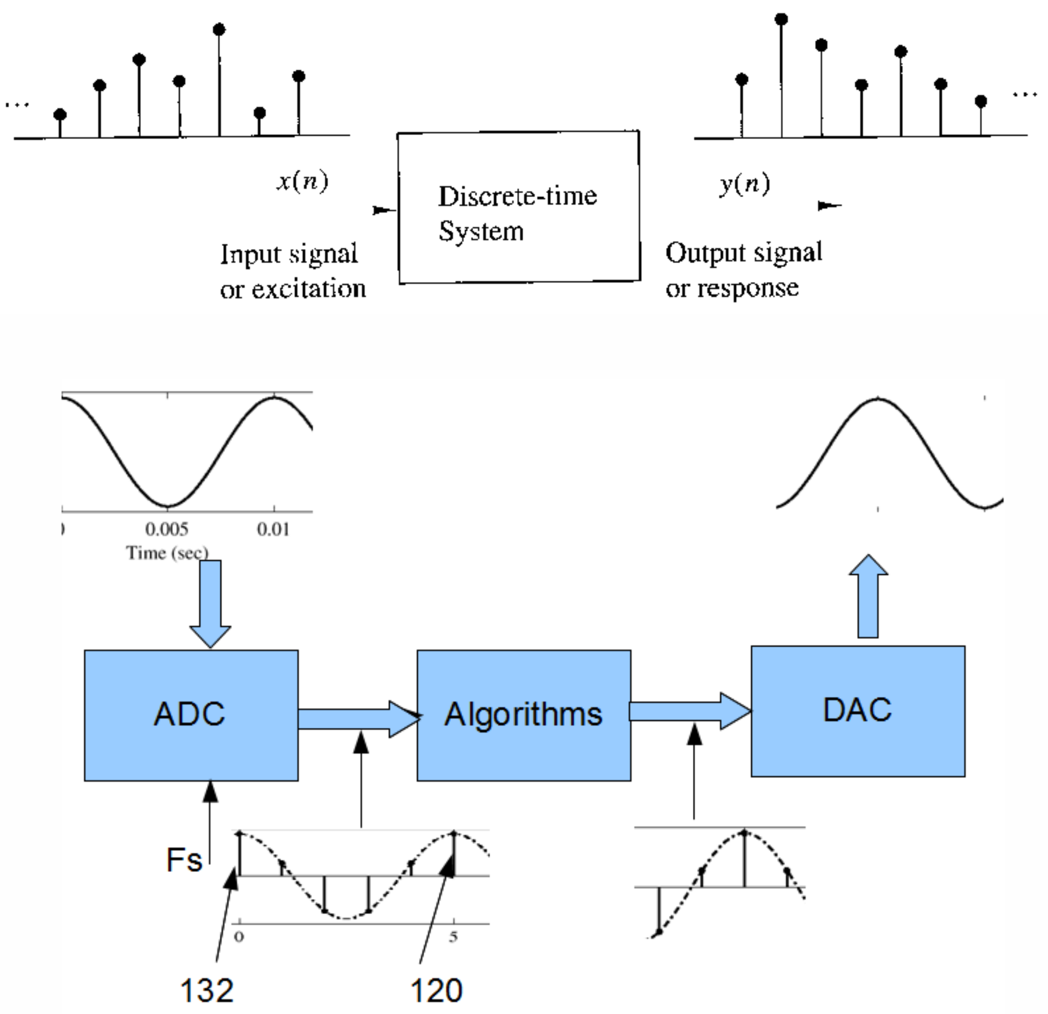
DSP system

- Analog input: (voltage, pressure, temperature, etc).
- Analog to digital converter (ADC) transform an analogue signal into digital numbers.
- The DSP-processor (computer or microprocessor) transform the signal, e.g. by filter it.
- Digital DSP-processor output is converted into an analogue signal by the Digital to Analog Converter (DAC).
- Analog output signal is filtered.



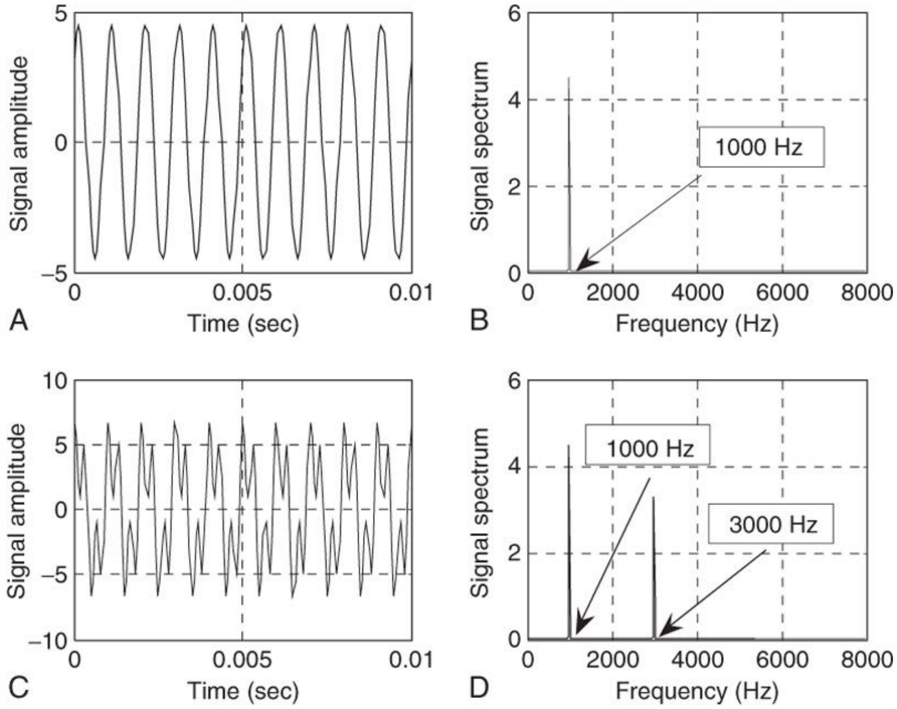
Discrete-Time Systems

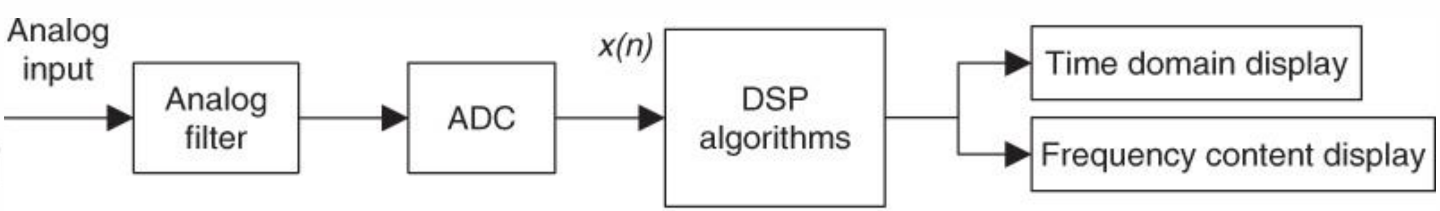
- The analogue input signal is sampled by the ADC
- The samples are processed by the discrete time system (i.e. transformed by some algorithms)
- The samples are converted to a continuous analogue signal by the DAC



Spectrum analysis

- **A spectrum analyse technique often used is Discrete Fourier Transforms (DFT).**
- **An efficient implementation of the DFT is the Fast Fourier Transform (FFT) algorithm.**



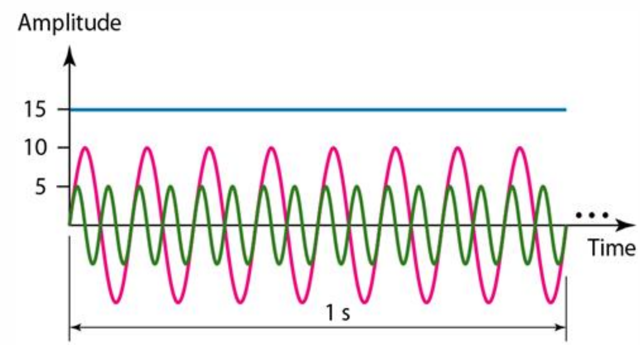


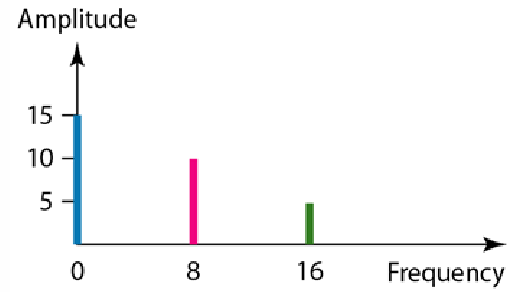
Time vs. frequency domain

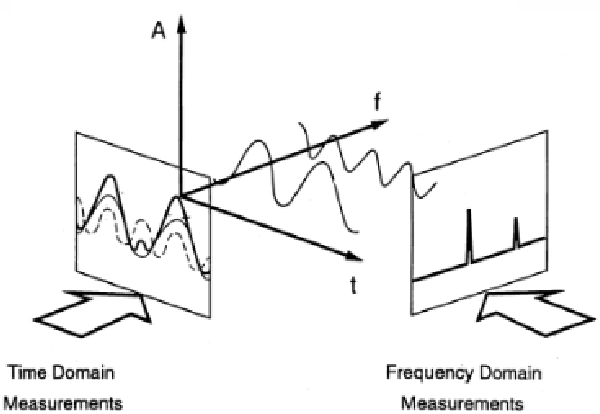
- **Time domain: representation of a signal with respect to time**

- **Frequency domain: representation of energy in a signal at each of many frequencies**

- **Time, frequency and amplitude can be illustrated in a tree dimension coordinate system**







Sampling Theorem

- **How often do we need to take a sample ?**
  - Low sampling rate saves memory ?
- **The lowest possible sampling rate is given by the SHANNON/NYQUIST Theorem**

Shannon Sampling Theorem

A continuous-time signal $x(t)$ with frequencies no higher than $f_{\max}$ can be reconstructed exactly from its samples $x[n] = x(nT_s)$, if the samples are taken at a rate $f_s = 1/T_s$ that is greater than $2f_{\max}$.

- **Half of the sampling frequency fs/2 is usually called the <u>Nyquist frequency</u>**

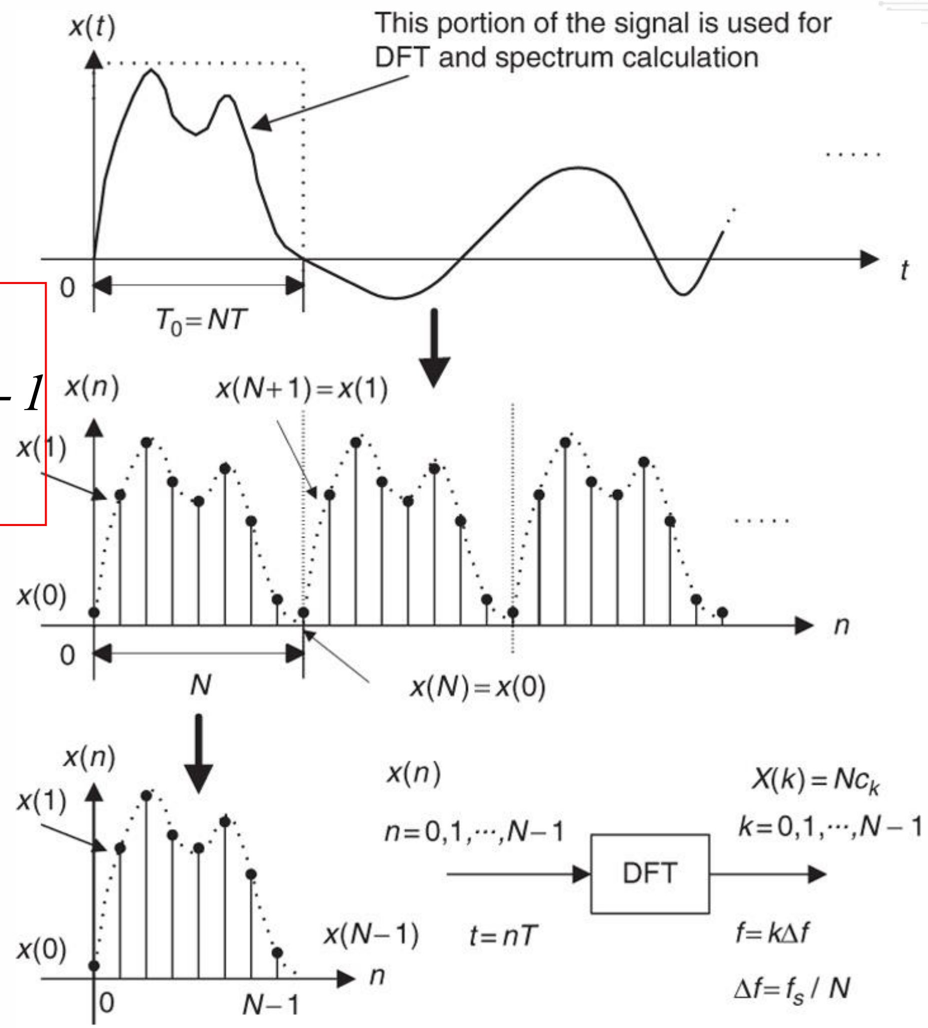
DFT formulas

- **The DFT formula is:**

$$X(k) = Nc_k = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi kn}{N}}, \quad k = 0,1,\dots,N-1$$

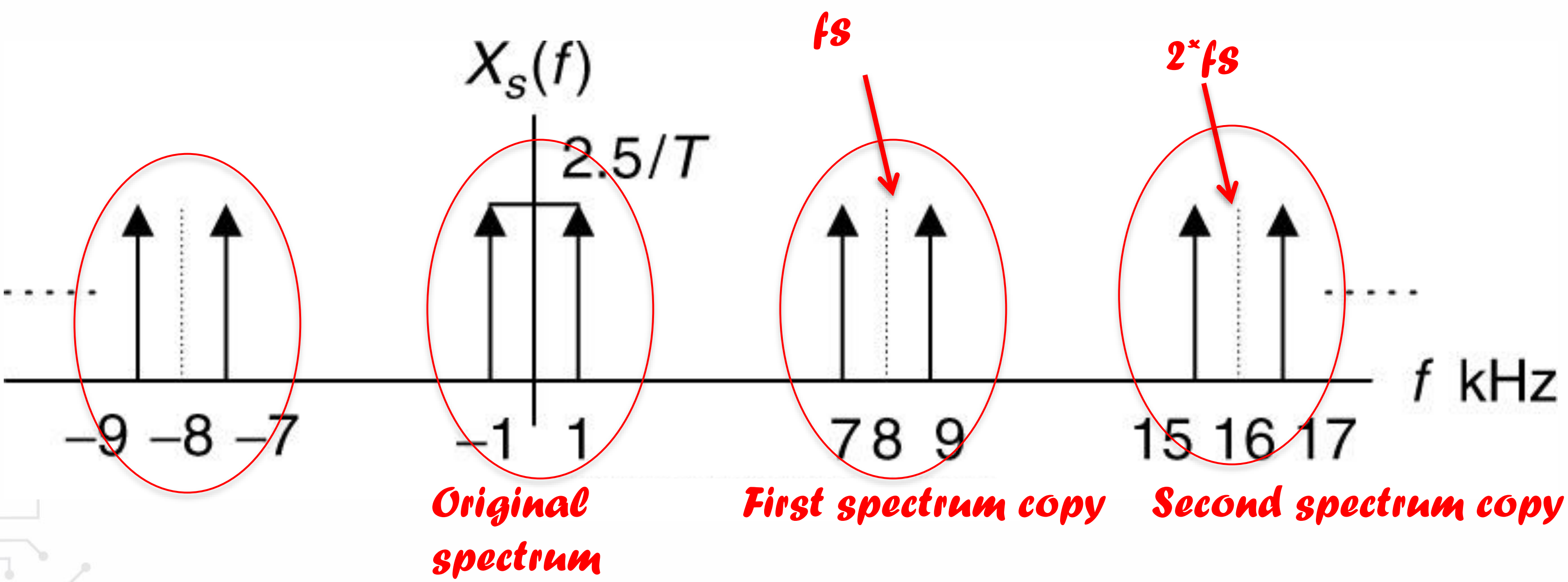
- **Inverse DFT or IDFT formula is:**

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j\frac{2\pi kn}{N}}, \quad n = 0,1,\dots,N-1$$



Spectrum duplicates, aliasing

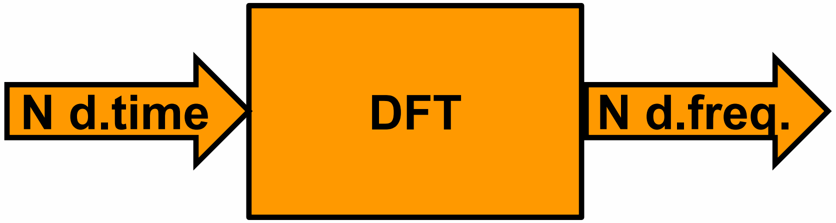
- A sampled spectrum copies itself to N*Fs, where N is any integer
- When these spectra overlaps aliasing occurs (aliasing is the “overlap”).



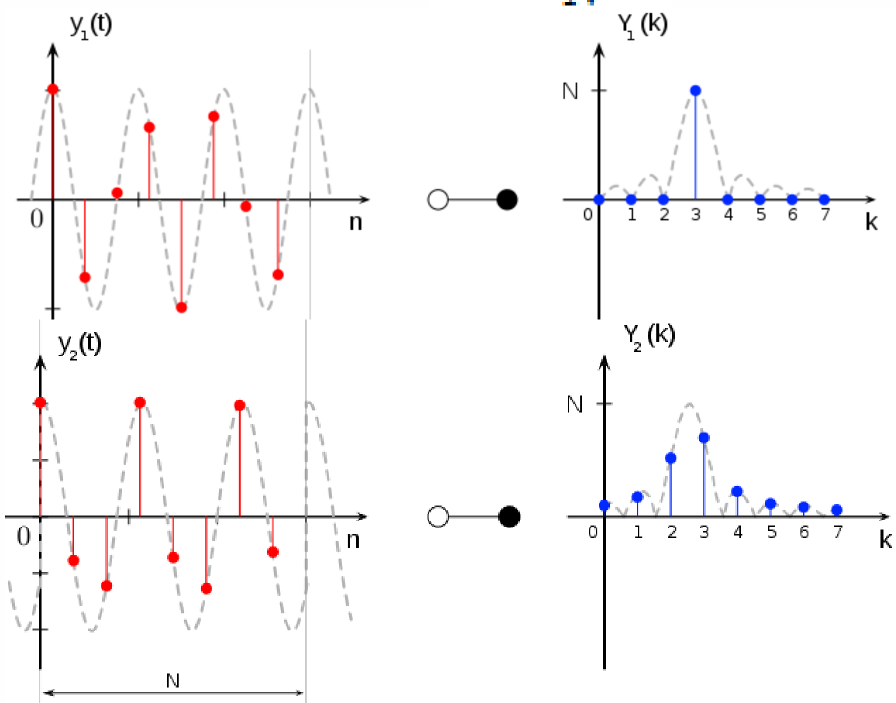
Spectral leakage

- Assume fs=1000 Hz and Fo=375 Hz, and 8 samples
- df=fs/N=1000/8=125 Hz, i.e. bin (k=3) fits (i.e. no spectral leakage)

- Assume fs=1000 Hz and Fo=312,5 Hz, and 8 samples
- df=fs/N=1000/8=125 Hz, i.e. the “true spectrum” (the DTFT spectrum) is at “k=2.5”, BUT, k is an integer, so bin 2 and 3 receives the power from it.
- Time domain windowing is equal to convolve with a sinc(x) in frequency domain.
- Ways to deal with this is windowing and / or synchronous sampling

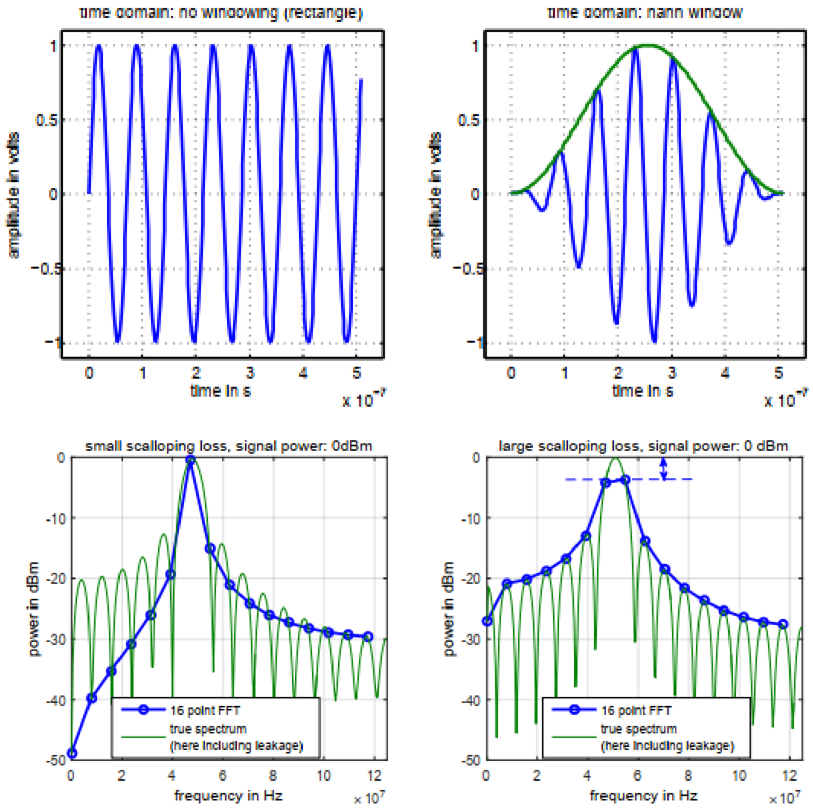


$$\Delta f = \frac{fs}{N}$$



Power estimation

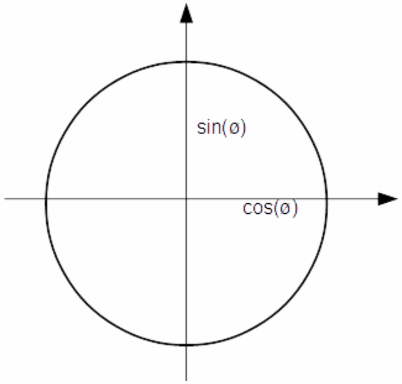
- **Coherent power gain.**
  - Amplitude reduction due to windowing.
- **Scalloping loss**
  - Sampling a signal may fall in between two bins.



Choosing the coil transmit frequency to e.g. Fs/4 simplifies the DFT…

DFT in low resource processors

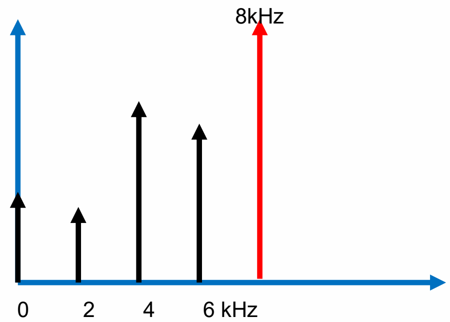
- **Let's assume the sampling frequency is set to 8 kHz, the Tx-coil frequency to 2 kHz and 4 samples (N=4)**



n=0	1	2	3	4	5
cos(ø)=1	0	-1	0	1	…
sin(ø)=0	1	0	-1	0	…

- **Consider if this can be used to implement the DFT, i.e. amplitude and phase calculations.**
- **Be aware of spectral leakage, i.e. N=4 is NOT enough samples**

$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k n}{N}}, \qquad \mathrm{k = 0,1,\ldots,N-1}$$



$$X(1) = \sum_{n=0}^{N-1} x[n] e^{-j\frac{\pi n}{2}}, \qquad \mathrm{k = 0,1,\ldots,N-1}$$

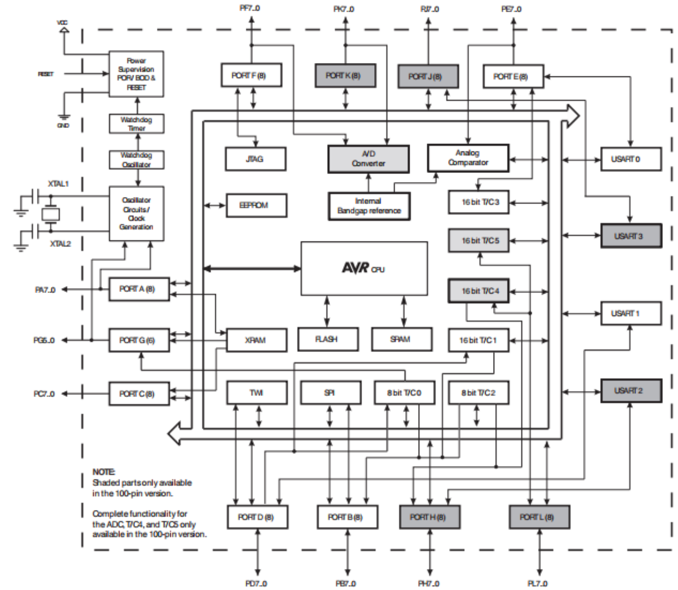
$$X(1) = \sum_{n=0}^{N-1} (x[n]\cos(\frac{\pi n}{2}) - jx[n]sin(\frac{\pi n}{2}))$$

$$X(1) = \sum_{n=0}^{N-1} x[n]\cos(\frac{\pi n}{2}) - j\sum_{n=0}^{N-1} x[n]sin(\frac{\pi n}{2})$$

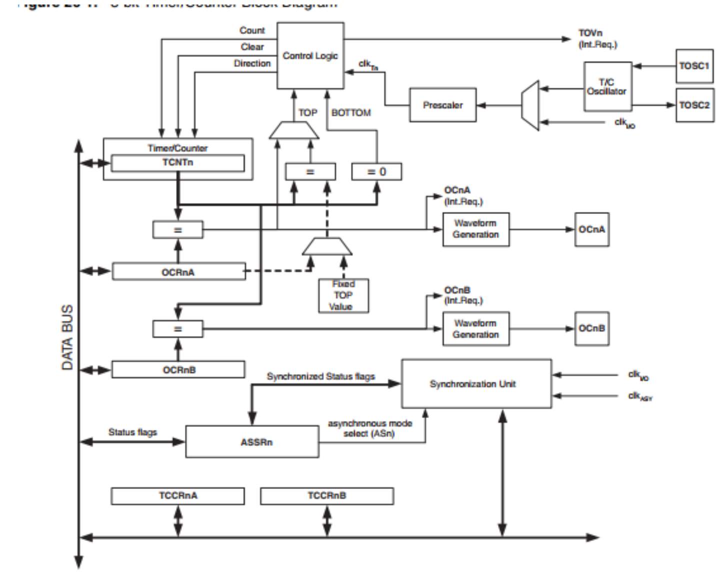
$$\mathrm{k = 0,1,\ldots,N-1}$$

Arduino software

- Arduino mega 2560 contains these blocks:
  - ADC
  - Timer interrupt



- E.g. timer 2 is can be used
  - Generate an interrupt whenever it overflows



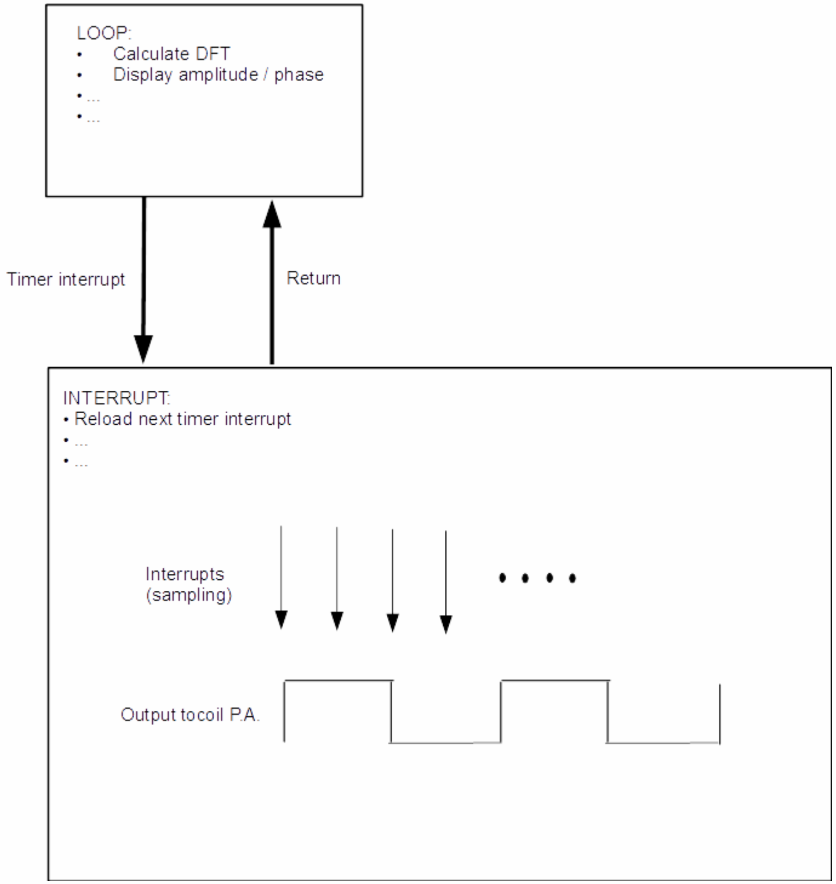
Arduino software

- To avoid scalloping loss, etc. synchronous sampling is needed. This can be implemented by using a timer interrupt in the processor (Arduino)

- Note, when <u>restarting the sampling, it must start at a rising edge of the output coil P.A. signal</u>.

- Be aware of possible race-conditions when using the samples in the LOOP and at the same time obtaining new ones in the INTERRUPT routine.

- Collect many samples (e.g. 64) before running the DFT. This reduces the impact of noise and spectral leakage in the results

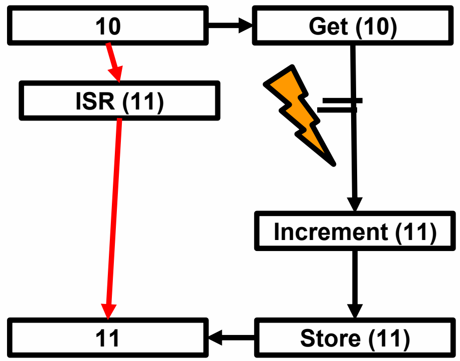


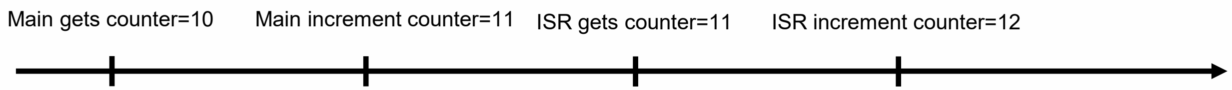
Race conditions…

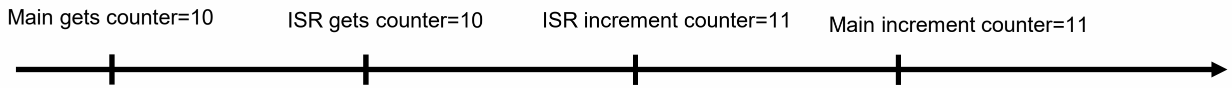
```
int counter = 0;

void main(void) {
    while (1) {
        counter++;   // non-atomic operation
    }
}

ISR(TIMER_INTERRUPT) {
    counter++;   // also non-atomic
}
```







Race conditions …

Disable Interrupts Around Critical Section

```
disable_interrupts();
counter++;
enable_interrupts();
```

Use Separate Variables

```
volatile int main_counter;
volatile int isr_counter;
```

volatile

```
volatile int counter;
```

Note: `volatile` prevents compiler optimization but **does NOT fix race conditions by itself**.

Arduino software

```
// You can also use an enum...
#define DO_SAMPLE   0
#define BUF_FULL    1
#define BUF_FREE    2


int stm=BUF_FREE;
int N=64;


 ISR(TIMER2_OVF_vect) {
    Reload timer
    Handle 2 kHz output (rising and falling edges)

    if(rising edge){
       PORTpin = HIGH;
       ...
       if(stm==BUF_FREE) stm=DO_SAMPLE;
       ...
    }
    if (stm==DO_SAMPLE){
       ADCVal[idx]= read a sample;
       if(++idx==N){
          idx=0;
          stm=BUF_FULL;
        }
    }
 }

void loop() {
if(stm==BUF_FULL){
    make DFT
    Filter results (magnitude and phase)
    Display magnitude and phase
    stm=BUF_FREE;
  }
}
```

Energy calculations …

Given Battery

- Capacity: **2000 mAh**
- Nominal voltage: **3.7 V** (typical Li-ion battery)

---

1 Convert mAh → Ah

$$2000 \text{ mAh} = 2 \text{ Ah}$$

---

2 Usage Time Example

Device Current Consumption

Assume the device draws:

- **200 mA**

Usage Time

$$\text{Time} = \frac{\text{Battery Capacity (mAh)}}{\text{Current (mA)}}$$

$$\text{Time} = \frac{2000}{200} = 10 \text{ hours}$$

✅ **Battery lasts ~10 hours**

3 Convert Battery Capacity → Energy (Watt-hours)

$$\text{Energy (Wh)} = \text{Voltage (V)} \times \text{Capacity (Ah)}$$

$$= 3.7 \times 2 = 7.4 \text{ Wh}$$

---

4 Convert Watt-hours → Joules

$$1 \text{ Wh} = 3600 \text{ J}$$

$$7.4 \times 3600 = 26{,}640 \text{ J}$$

✅ **Battery energy = 26.6 kJ**

---

5 Power-Based Usage Example

Device Power Consumption

Assume the device consumes:

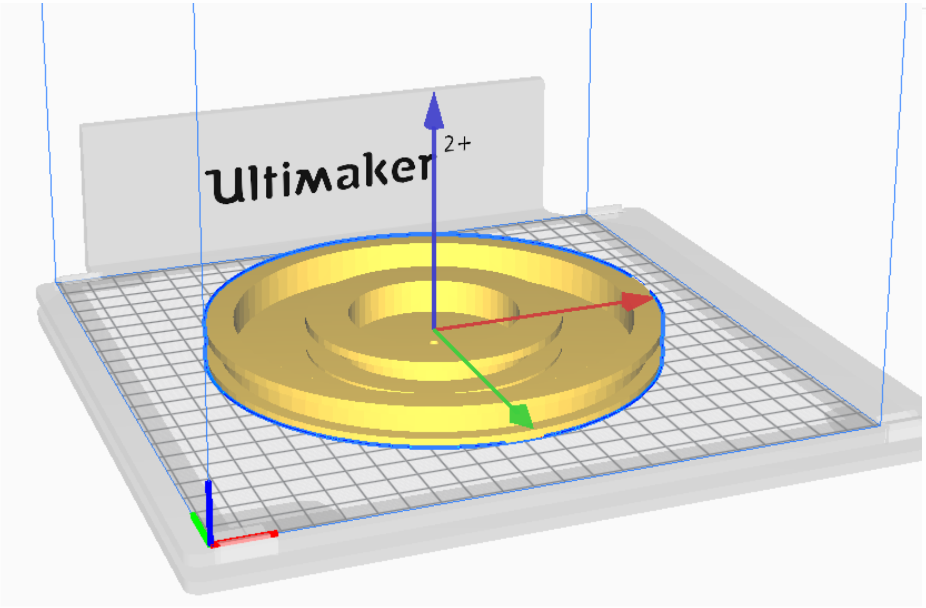
- **1 W**

Usage Time

$$\text{Time} = \frac{\text{Energy (Wh)}}{\text{Power (W)}}$$

$$= \frac{7.4}{1} = 7.4 \text{ hours}$$

Coils



- Coil-form – you chose it, i.e. freely chose how the coils are suspended…

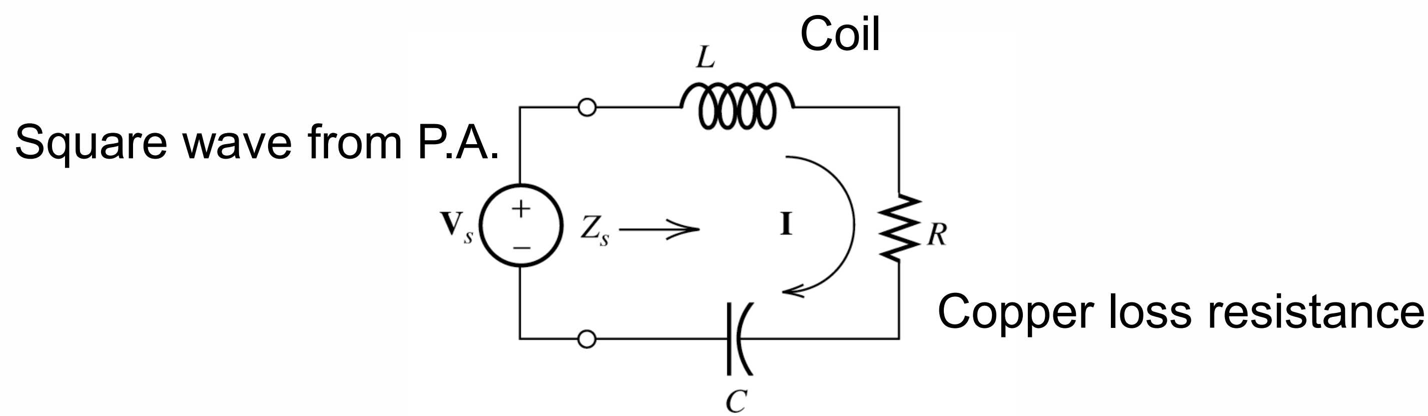
- WARNING coils can be dangerous…

Power / tx-coil

- **To maximize the sensitivity of the metal-detector the magnetic flux density (B-field) must be maximized:**

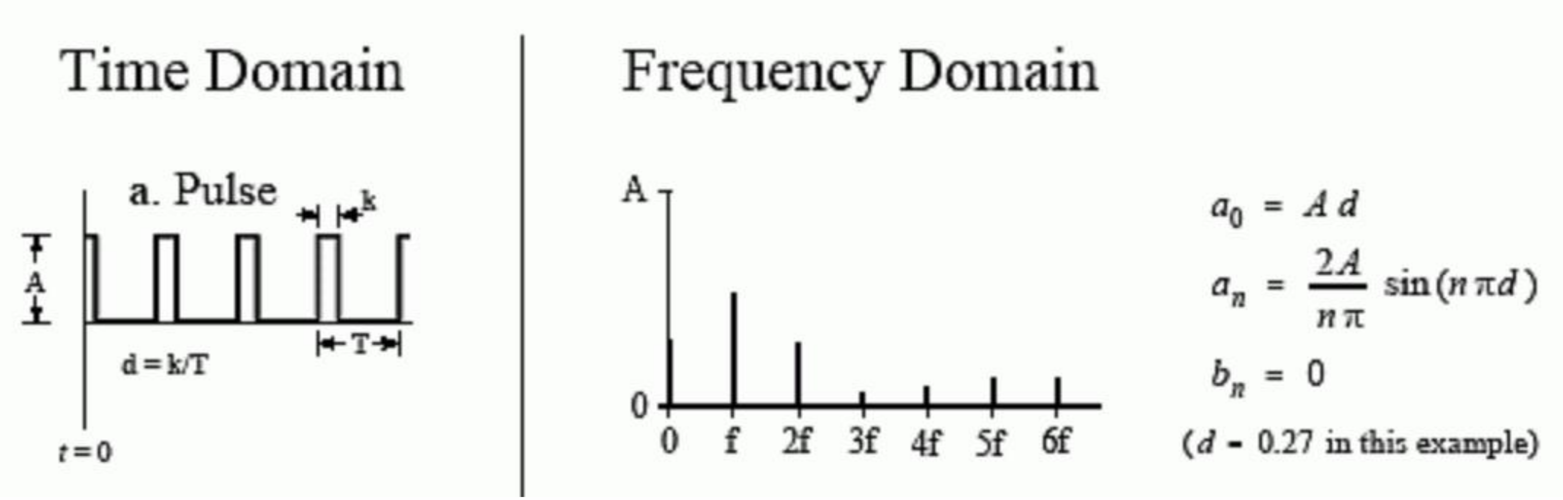
$$|B| = \frac{\mu 0 \cdot I \cdot N}{2 \cdot r}$$

  - where: I is current, N is coil-turns, r is coil-radius
- **Calculate the <u>energy</u> available from the battery and find the amount available for the transmit coil – this energy is then converted into a coil current.**
- **Tank circuit:**



Capacitor than makes the circuit resonate at the Transmit frequency (at resonance Z_s=R)

Tx-coil signals



- **The tank circuit is driven by a square-wave from the P.A. stage; however, at resonance this square-wave can be approximated by the Fundamental Frequency (FF).**
- **The Fourier-series coefficient of the FF is:** $$a_n = 2\frac{A}{n\pi}\sin\left(\frac{n\pi}{2}\right)$$
  - where A is the square-wave amplitude, n is 1 (fundamental frequency).
  - **Note that a_n is the peak value of the sine; thus dividing it with the square root of 2 yields the RMS voltage of the FF.**

Power / tx-coil

- **Knowing that the tank circuit is pure resistive (R), at resonance, the RMS voltage of the FF, and the requirements for the RMS coil-current, the value of R can be found.**
  - Remember to compensate the FF-voltage with some losses from the P.A. stage
- **Because the VLF frequency is very low the current is considered to be conducted equally through the area of the copper-wire (i.e. the skin effect is ignored). Hence, the number of turns in the transmit-coil can be found:**

$$Ntx = \frac{R \cdot rt^2}{2 \cdot r \cdot \rho_c}$$

  - Where rt is radius of the copper-thread, ρ_c is copper resistivity
  - Coil radius (r) must be chosen

Power / tx-coil

- **Calculate the self-inductance from:** $L = \dfrac{0.394\, r^2\, N^2}{9r + 10A}$

NOTE:

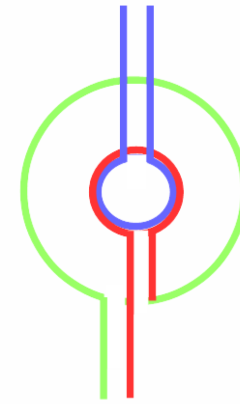
L = inductance (in microhenries)
r = radius of coil (in cm)
N = number of turns
A = Height of the coil (in cm)

- **NOTE "A" is the hight of the coil approximately 1 cm in many cases…**
- **Calculate the capacitor (C) from the LC-tank resonance equation.**

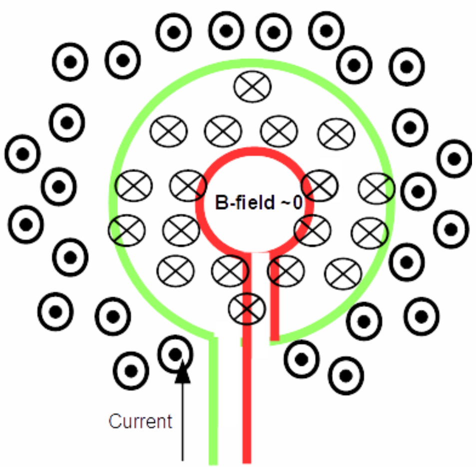
- **Calculate the modulus value of the B-field (chapter 5 in "Fundamentals of applied electromagnetics", seventh edition, Fawwaz T. et. al.)**

Coils

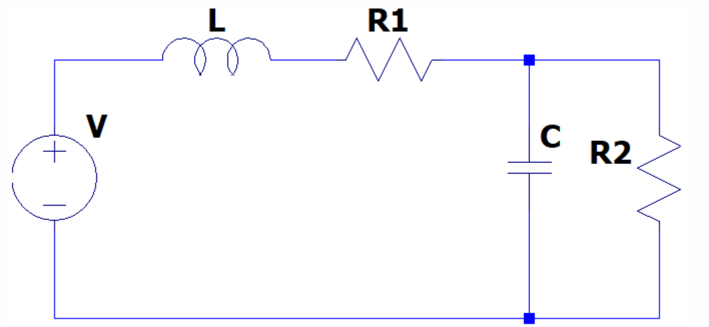
- **Example with three coils:**
  - Green is Tx coil
  - Red is feedback coil
  - Blue is Rx coil
- **The feedback coil balance out the field from the Tx coil inside the Rx-coil**



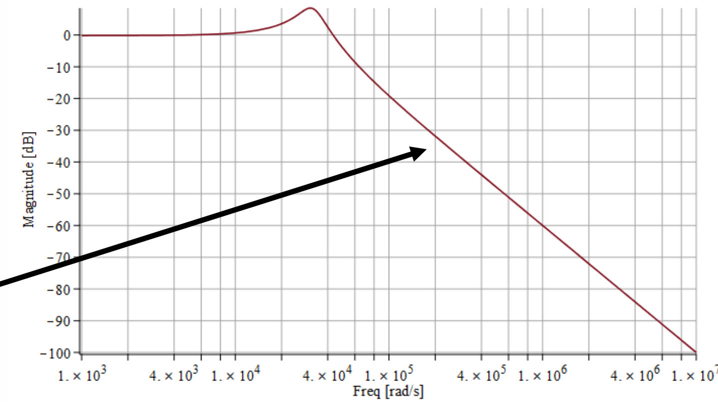
- **Calculate the value of the H or B-field in the wound solenoids (chapter 5 in "Fundamentals of applied electromagnetics", seventh edition, Fawwaz T. et. al.)**
- **Assumptions:**
  - The field is homogeneous inside the Tx-coil.
  - The Tx-current is passing through the feedback-coil.
- **Calculate the number of turns for a given radius for the feedback-coil**
  - This is only an approximation – you need to measure and fine-tune in the lab
- **To obtain good sensitivity the inductance of the receiver coil (Rx) should be larger than 10 mHy.**

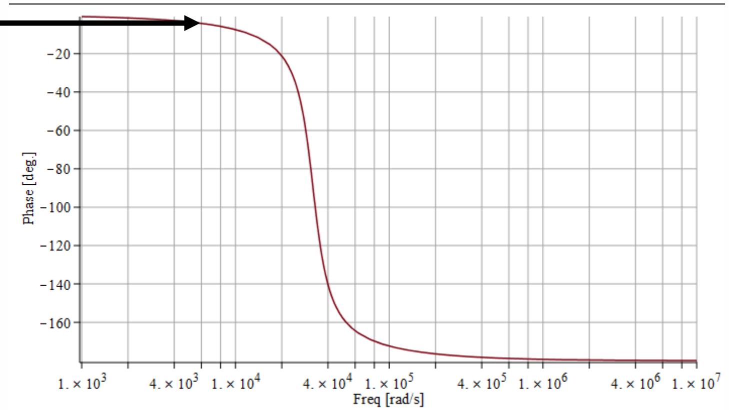


Receiver noise filtering

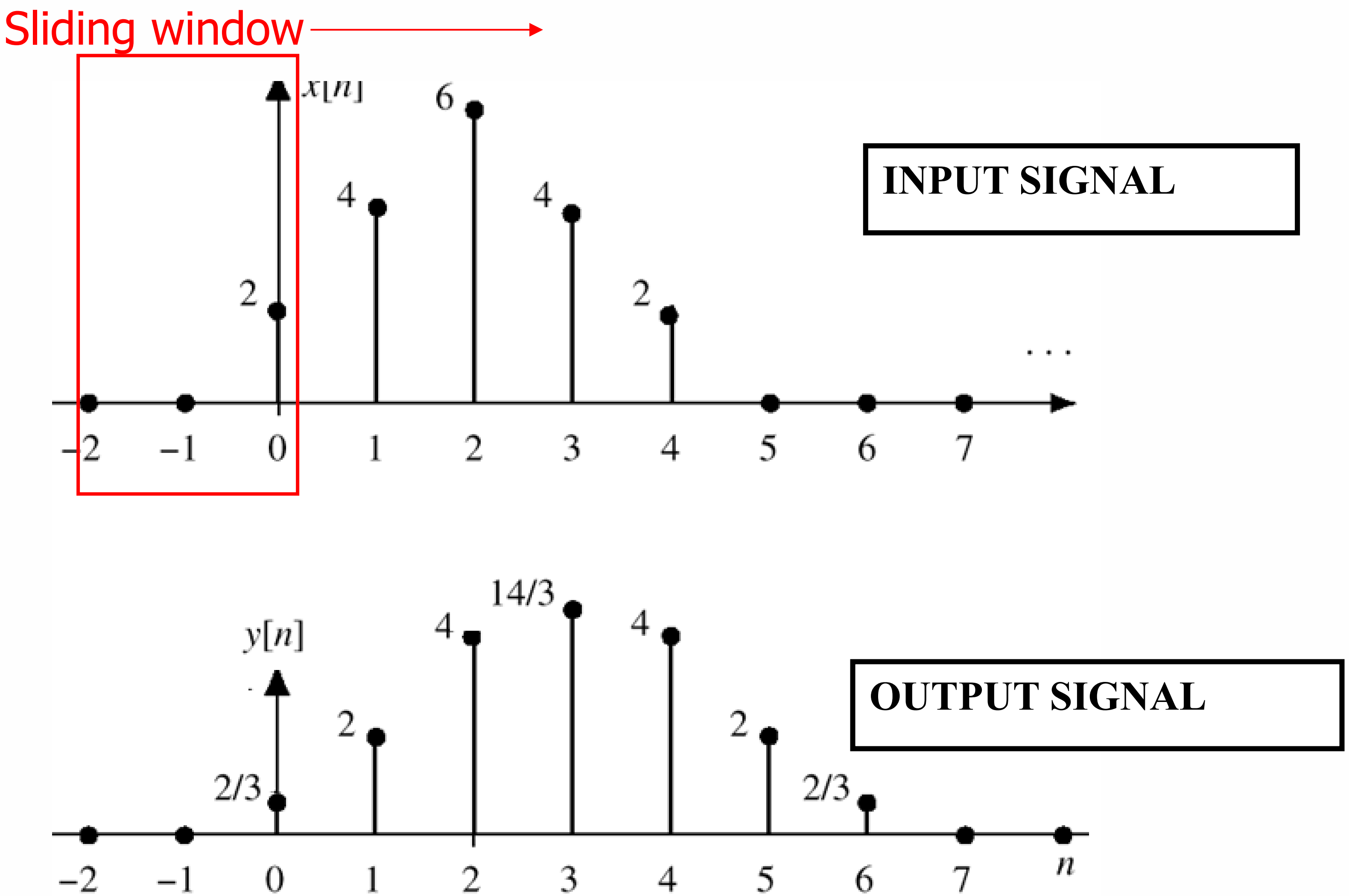


- **Assume:**
  - V is the voltage generated by the mutual coupling to a detected object
  - Voltage across R2 is the input to the Rx amplifier, i.e. R2 is the input impedance of the amplifier.
  - L is the inductance in the receive coil
  - R1 is the "ohmic resistance" in coil L
  - C is an applied filter capacitor
- **To find the best component values, you need to consider:**
  - Needed damping of high-frequency noise
  - Unwanted phase changes caused by variations (jitter) of the Tx-frequency





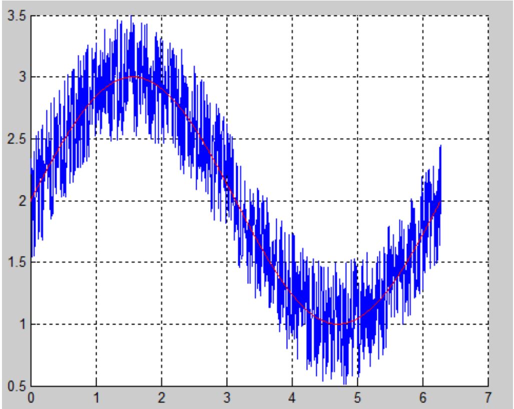
Filtering display information

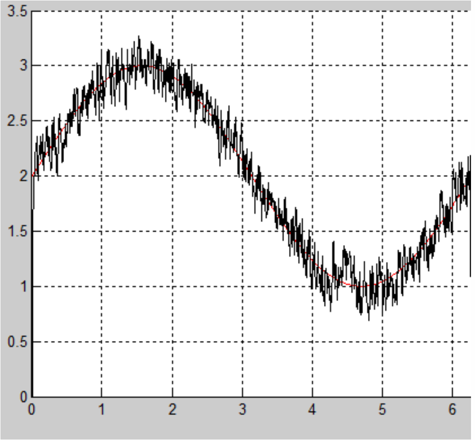


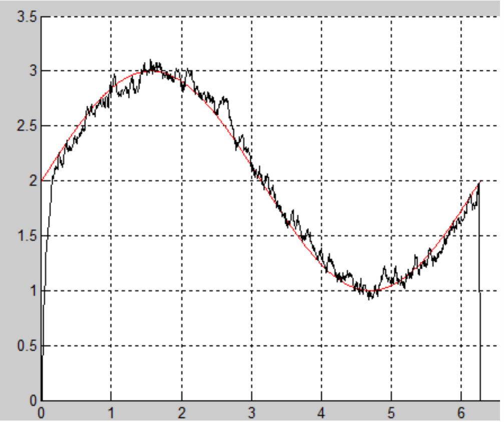
$$y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n-2])$$

Filtering display information

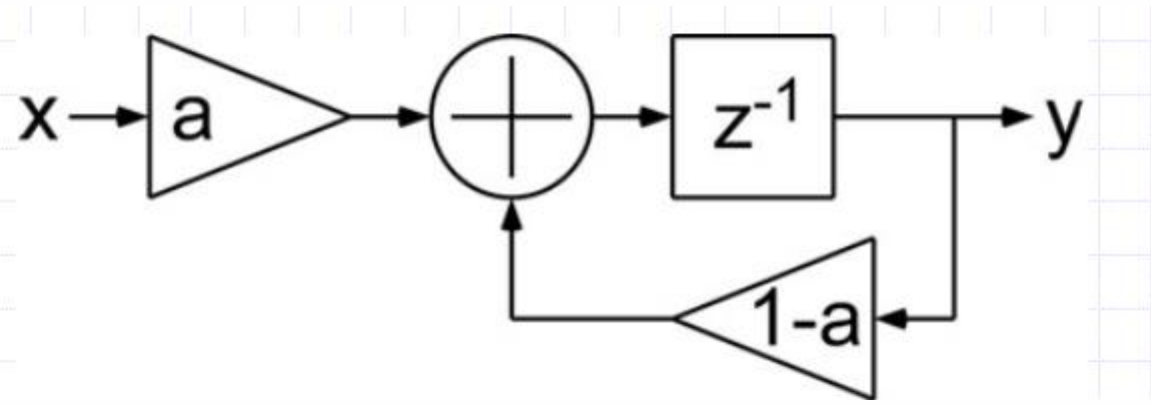
- **A recursive average low-pass filter**



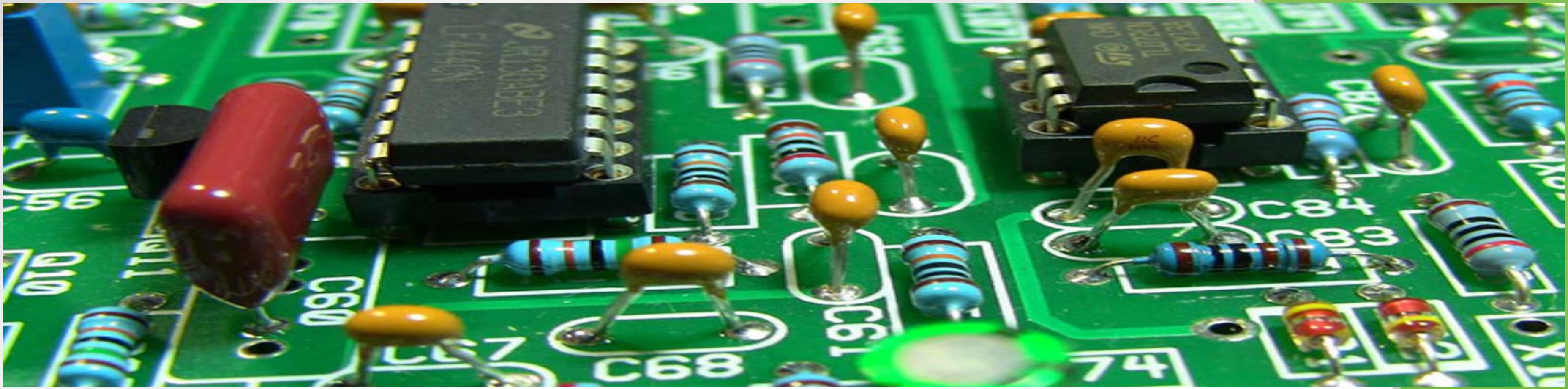




$$acc = acc * 0.6 + 0.4 * smpl \qquad acc = acc * 0.9 + 0.1 * smpl$$



Example: Metaldetector video



Thank you for your attention !

Questions ?